

Elizabeth Sang Nguyen

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EDUCATION

California Institute of Technology (Caltech)

Bachelor of Science in Computer Science

Pasadena, CA

2025 - 2029

GPA: 4.1/4.0

Relevant Coursework: Computer Programming, Data Structures & Algorithms, Software Design, Computational Science & Engineering, Multivariable Calculus, Linear Algebra

Activities: First Year Research (FSRI), Swim & Dive, HackTech Organizer, Society of Women Engineers

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, C, C#, MATLAB

Frameworks & Tools: Ursina, Unity, Figma, Git, GitHub, NumPy, PyTorch, Claude Code

Concepts: Algorithm Design & Analysis, Data Structures, Object-Oriented Programming, Machine Learning

EXPERIENCE

NASA Jet Propulsion Laboratory (JPL)

Machine Learning Intern

Pasadena, CA

June-August 2026

- Developed a deep learning diagnostic system using convolutional neural networks to analyze Monte Carlo Markov Chain (MCMC) corner plots for exoplanet parameter inference.
- Built an AI-powered agent enabling natural language querying across 14M+ exoplanet spectroscopy observations from JWST and HST telescopes.
- Supporting data processing for the upcoming [ARIEL](#) space telescope mission.

California Institute of Technology (Caltech)

Undergraduate Researcher

Pasadena, CA

July–September 2025

- Redesigned and implemented interactive SQL-based instructional datasets and tooling for an upper-level databases curriculum, emphasizing applied data exploration and visualization.
- Conducted quantitative benchmarking across three peer institutions to inform curriculum modernization.

Microsoft

Discovery Intern

Redmond, WA

July–August 2025

- 1 of 40 Washington-wide high school seniors selected for highly competitive Microsoft internship.
- Conducted user research and prototyping to inform UI needs for 1.63M+ active users of Viva Glint.
- Proposed feature changes to increase user comment quality; adopted into FY26 product backlog.
- Designed front-end Copilot interface aligned with Microsoft accessibility guidelines.

Code Ninjas

Coding Instructor

Newcastle, WA

August 2024–April 2025

- Taught JavaScript, Unity, and robotics to 30+ children (ages 4–13).
- Designed weekly curricula covering fundamental concepts such as conditionals, loops, and debugging.

PROJECTS

Hand Gesture Recognition — Python, MediaPipe, OpenCV

- Building a real-time finger-tracking system with MediaPipe hand landmark detection
- In progress: developing touchless virtual flute using computer vision and audio signal processing, mapping real-time hand landmark tracking to flute fingering and microphone-based breath detection for note onset

House Customizer — Python (Ursina Engine)

- Designed a modular UI system with unified cycling interface for dynamic component updates.
- Built scalable state management patterns linking categories, assets, and UI elements.
- Leveraged event-driven programming for efficient part selection and texture cycling.